

MH2500 Probability and Introduction to Statistics

Slides for Week 12-1

Topics: MGF continued

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Theorem

If X has the mgf $M_X(t)$ and $Y = a + bX$, then Y has mgf

$$M_Y(t) = e^{at} M_X(bt).$$

Proof:

$$\begin{aligned} M_Y(t) &= \mathbb{E}(e^{tY}) \\ &= \mathbb{E}(e^{t(a+bX)}) \\ &= \mathbb{E}(e^{at+btX}) \\ &= \mathbb{E}(e^{at} e^{btX}) \\ &= e^{at} \mathbb{E}(e^{btX}) \\ &= e^{at} M_X(bt). \end{aligned}$$

MGF of General Normal Distribution

From the earlier example, for $Z \sim N(0, 1)$, $M_Z(t) = e^{t^2/2}$.

Let $X \sim N(\mu, \sigma^2)$. Then $X = \mu + \sigma Z$ and

$$M_X(t) = e^{\mu t} M_Z(\sigma t) = e^{\mu t} e^{\sigma^2 t^2/2} = e^{\mu t + \sigma^2 t^2/2}.$$

By **differentiating**, we have

$$M'(t) = (\mu + \sigma^2 t) e^{\mu t + \sigma^2 t^2/2},$$

$$M''(t) = (\sigma^2) e^{\mu t + \sigma^2 t^2/2} + (\mu + \sigma^2 t)^2 e^{\mu t + \sigma^2 t^2/2}.$$

This gives

$$\mathbb{E}(X) = M'(0) = \mu, \quad \mathbb{E}(X^2) = M''(0) = \sigma^2 + \mu^2,$$

and hence $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sigma^2$.

Theorem

If X and Y are independent random variables with mgf's M_X and M_Y and $Z = X + Y$, then $M_Z(t) = M_X(t)M_Y(t)$ on the common interval where both mgf's exist.

Proof:

$$M_Z(t) = \mathbb{E}(e^{tZ}) = \mathbb{E}(e^{t(X+Y)}) = \mathbb{E}(e^{tX}e^{tY})$$

by the assumption of independence,

$$= \mathbb{E}(e^{tX}) \mathbb{E}(e^{tY}) = M_X(t)M_Y(t).$$

Sum of independent Poisson variables

Suppose X is Poisson with parameter λ and Y is Poisson with parameter μ , and that X and Y are independent random variables. Then $X + Y$ is Poisson with parameter $\mu + \lambda$.

We know that X and Y have mgf

$$M_X(t) = e^{\lambda(e^t-1)} \quad M_Y(t) = e^{\mu(e^t-1)}.$$

Hence

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{(\lambda+\mu)(e^t-1)}.$$

Therefore, $X + Y$ is Poisson with parameter $\lambda + \mu$.

Sum of independent Normal variables

Suppose $X \sim N(\mu, \sigma^2)$ and $Y \sim N(\nu, \tau^2)$ and are independent of each other.

- Find the mgf of $X + Y$.

Solution:

$$\begin{aligned} M_{X+Y}(t) &= M_X(t)M_Y(t) = \left(e^{\mu t + \sigma^2 t^2/2}\right) \left(e^{\nu t + \tau^2 t^2/2}\right) \\ &= e^{(\mu+\nu)t + (\sigma^2 + \tau^2)t^2/2}. \end{aligned}$$

Thus $X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2)$.

- ① If X and Y have a joint distribution, their **joint mgf** is defined as

$$M_{XY}(s, t) = \mathbb{E}(e^{sX+tY}).$$

- ② It can be shown that **two random variables X and Y are independent if and only if their joint mgf factors into the product of the mgf's of the marginal distributions.**
- ③ A major limitation of the mgf is that it may not exist.